DATA SET DS1201

21. How Long Is It? How good are you at estimating? To test a subject's ability to estimate sizes, he was shown 10 different objects and asked to estimate their length or diameter. The object was then measured, and the results were recorded in the table below.

Object	Estimated (centimeters)	Actual (centimeters)
Pencil	17.7	15.2
Dinner plate	24.1	26.0
Book 1	19.0	17.1
Cell phone	10.1	10.7
Photograph	36.8	40.0
Toy	9.5	12.7
Belt	106.6	105.4
Clothespin	6.9	9.5
Book 2	25.4	23.4
Calculator	8.8	12.0

a. Find the least-squares regression line for predicting the actual measurement as a function of the estimated measurement.

b. Plot the points and the fitted line. Does the assumption of a linear relationship appear to be reasonable?

Q21

$n = 10$

a) $\Sigma x = 264.9$ $\Sigma x^2 = 14935.37$ $\bar{x} = 26.49$

$\Sigma y = 272$ $\Sigma y^2 = 14966.2$ $\bar{y} = 27.2$

$\Sigma xy = 14922.41$

$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 14935.37 - \frac{264.9^2}{10} = 7018.169$

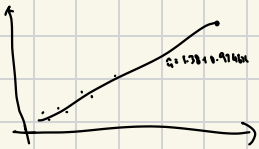
$S_{yy} = \Sigma y^2 - \frac{(\Sigma y)^2}{n} = 14966.2 - \frac{272}{10} = 7367.8$

$S_{xy} = \Sigma xy - \frac{\Sigma x \Sigma y}{n} = 14922.41 - \frac{264.9 \times 272}{10} = 7717.13$

$b = \frac{S_{xy}}{S_{xx}} = \frac{7717.13}{7018.169} = 0.9946$ $a = \bar{y} - b(\bar{x}) = 27.2 - 0.9946 \times 26.49 = 1.38$

$\hat{y} = 1.38 + 0.9946x$

b)



It is reasonable.

DATA SET DS1202

22. Armspan and Height Leonardo da Vinci (1452–1519) drew a sketch of a man, indicating that a person's armspan (measuring across the back with your arms outstretched to make a "T") is roughly equal to the person's height. To test this claim, we measured eight people with the following results:

Person	1	2	3	4
Armspan (centimeters)	172	158	165	176
Height (centimeters)	175	157	165	177

Person	5	6	7	8
Armspan (centimeters)	172	175	157	153
Height (centimeters)	170	170	160	157

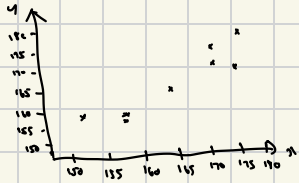
a. Draw a scatterplot for armspan and height. Use the same scale on both the horizontal and vertical axes. Describe the relationship between the two variables.

b. If da Vinci is correct, and a person's armspan is roughly the same as the person's height, what should the slope of the regression line be?

c. Calculate the regression line for predicting height based on a person's armspan. Does the value of the slope b confirm your conclusions in part b?

d. If a person has an armspan of 157 centimeters, what would you predict the person's height to be?

Q22 a)



b) It should be linearly increasing

c) $n = 8$

$\Sigma x = 1328$ $\Sigma x^2 = 221016$ $\bar{x} = 166$

$\Sigma y = 1331$ $\Sigma y^2 = 221877$ $\bar{y} = 166.375$

$\Sigma xy = 221414$

$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 568$

$S_{yy} = \Sigma y^2 - \frac{(\Sigma y)^2}{n} = 431.875$

$S_{xy} = \Sigma xy - \frac{\Sigma x \Sigma y}{n} = 468$

$b = \frac{S_{xy}}{S_{xx}} = 0.824$ $a = \bar{y} - b(\bar{x}) = 29.6$

$\hat{y} = 29.6 + 0.824x$

d) Yes

$y = 29.61 + 0.824(157) = 159$ cm

DATA SET DS1204

10. Body Mass Index A study using body mass index (BMI)—an index of obesity—as a function of income ($ thousands) reported the following data for California in 2016.[2]

Income	15	20.5	30	40	60	75
BMI	31.2	29.3	27.4	27.3	26.8	20.0

a. If the researcher thinks that BMI is a function of income, which of the two variables is the independent variable x and which is the dependent variable y?

b. Find the least-squares line relating BMI to income.

c. Construct the ANOVA table for the linear regression.

Q10 a) x: income y: BMI

b) $\Sigma x = 240.5$ $\bar{x} = 40.083$ $\Sigma x^2 = 12370.25$

$\Sigma y = 162$ $\bar{y} = 27$ $\Sigma y^2 = 4446.22$

$\Sigma xy = 6040.65$

$S_{xx} = 2730.208$

$S_{xy} = 72.22$

$S_{yy} = -492.85$

$b = -0.04755$ $a = 32.9$

$\hat{y} = 32.9 - 0.04755x$

c)

Source	df	SS	MS	F
Regression	1	59.44	59.44	18.6
Error	6-1-1=4	12.78	3.194	
Total	6-1=5	72.22		

11. Professor Asimov Professor Isaac Asimov wrote nearly 500 books during a 40-year career. In fact, as his career progressed, he became even more productive in terms of the number of books written within a given period of time.[3] The data give the time in months required to write his books in increments of 100:

Number of Books, x	100	200	300	400	490
Time in Months, y	237	350	419	465	507

a. Assume that the number of books x and the time in months y are linearly related. Find the least-squares line relating y to x.

b. Plot the time as a function of the number of books written using a scatterplot, and graph the least-squares line on the same paper. Does it seem to provide a good fit to the data points?

c. Construct the ANOVA table for the linear regression.

Q11 a) $\sum x = 1490$ $\bar{x} = 298$ $\sum x^2 = 540100$

$\sum y = 1978$ $\bar{y} = 395.6$ $\sum y^2 = 827504$

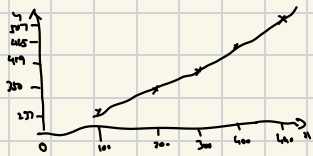
$\sum xy = 653830$

$S_{xx} = 96080$ $S_{yy} = 45007.2$ $S_{xy} = 64386$

$b = 0.67$ $a = 195.9$

$\hat{y} = 195.9 + 0.67x_i$

b)



c)

Total SS: $S_{yy} = 45007.2$

SSR: 43146.93

SSE: $45007.2 - 43146.93 = 1860.27$

Source	df	SS	MS	F
regression	1	43146.93	43146.93	69.58
error	3	1860.27	620.09	
total	4	45007.2		

9. Lots of Highways The number of miles of U.S. urban roadways (millions of miles) for the years 2000–2015 is reported below.[6] The years are simplified as years 0 through 15.

DATA SET DS1210

Miles of Roadways (millions)	0.85	0.88	0.89	0.94	0.98	1.01	1.03	1.04
Year − 2000	0	1	2	3	4	5	6	7

Miles of Roadways (millions)	1.07	1.08	1.09	1.10	1.11	1.18	1.20	1.21
Year − 2000	8	9	10	11	12	13	14	15

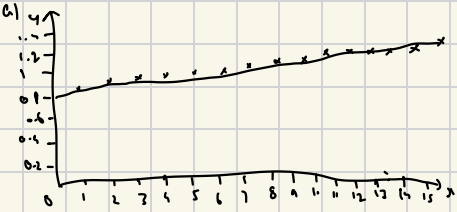
a. Draw a scatterplot of the number of miles of roadways in the U.S. over time. Describe the pattern that you see.

b. Find the least-squares line describing these data. Do the data indicate that there is a linear relationship between the number of miles of roadways and the year? Test using a t statistic with $\alpha = .05$.

c. Construct the ANOVA table and use the F statistic to answer the question in part b. Verify that the square of the t statistic in part b is equal to F.

d. Calculate r^2. What does this value tell you about the effectiveness of the linear regression analysis?

Q9 a)



b) $S_{xx} = 340$ $S_{yy} = 0.186375$ $S_{xy} = 7.85$

$b = 0.023$ $a = 0.868$

Total SS: 0.186375 SSR: 0.1812 SSE: 0.0051 MSR: 0.1812 MSE: 0.0003366 F: 538.39

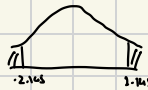
① $H_0: \beta = 0$ $H_1: \beta \neq 0$

② $\alpha = 0.05$

③ $t_{stat} = \frac{b - 0}{\sqrt{MSE/S_{xx}}} \sim t_{n-2}$

④ $t^* = \frac{0.023}{\sqrt{0.0003366/340}} = 22.27$

⑤ RR



Since t^* lies within the rejection region, we reject H_0.

d) $r^2 = \frac{SSR}{SST} = \frac{0.1812}{0.186375} = 0.972$

The model is 97.2% useful.

c)

Source	df	SS	MS	F
reg	1	0.1812	0.1812	499.29
error	4	0.0051	0.0003366	
total	5	0.186375		

① $H_0: \beta = 0$ $H_1: \beta \neq 0$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{MSR}{MSE} \sim F_{(1,n-2)}$

④ R.R. $F_{1,n;0.05} = 4.6$

Since $F^* > F_{1,14;0.05}$, rej H_0.

Model is useful.

DATA SET BS1211

10. Recidivism Recidivism refers to the return to prison of a prisoner who has been released or paroled. The data that follow reports the group median age at which a prisoner was released from a federal prison and the percentage of those arrested for another crime.[7] Use the *MS Excel* printout to answer the questions that follow.

Group Median Age (x)	22	27	32	37	42	47	52
% Arrested (y)	64.7	59.3	52.9	48.6	44.5	37.7	23.5

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9779
R Square	0.9564
Adjusted R Square	0.9477
Standard Error	3.1622
Observations	7.0000

ANOVA

	df	*SS*	*MS*	*F*	Significance *F*
Regression	1	1096.251	1096.251	109.631	0.000
Residual	5	49.997	9.999		
Total	6	1146.249			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	93.617	4.581	20.436	0.000	81.842	105.393
x	−1.251	0.120	−10.471	0.000	−1.559	–

a. Find the least-squares line relating the percentage arrested to the group median age.

b. Do the data provide sufficient evidence to indicate that x and y are linearly related? Test using the t statistic at the 5% level of significance.

c. Construct a 95% confidence interval for the slope of the line.

d. Find the coefficient of determination and interpret its significance.

(Q10) a) $a = 93.617$ $b = -1.251$

$\hat{y} = 93.617 - 1.251x$

b) $H_0: \beta = 0$ $H_1: \beta \neq 0$

$\alpha = 0.05$

$t_{STAT} = \frac{b - 0}{\sqrt{\frac{MSE}{S_{xx}}}} \sim t_{n-2}$

$t^* = -10.471$

RR $t_{5;0.025} = -2.571$

Since $t^* < t_{5;0.025}$, reject H_0.

c) $b_1 \; t_{5;0.025}\sqrt{\frac{MSE}{S_{xx}}} = -1.251 \pm (-2.571)\left(\sqrt{\frac{9.999}{700}}\right)$

$= [-0.942, -1.559]$

d) $R^2 = \frac{SSR}{SST} = \frac{1096.251}{1146.249} = 0.95678$

The model is 95.68% useful.